



Cambrian College

School of Business and Information Technology, Creative Arts, Design, Music and Hospitality

Course Outline

Course Title	Enterprise Systems				
Course Code:	SYS1004	Credit Value:	3	Credit Hours:	42
Programs:	BAGC I.T. Business Analysis BTHP I.T. Business Analysis - Hanson				
Equivalencies		Prerequisites		Corequisites	

This course may be delivered in a variety of different formats: 100% in-class, 100% online (or a blend of both), videoconferencing, distributed learning or off-campus. Please confirm with your faculty member which format will be used for your section of this course.

General Education Course: ☐

Eligible for PLAR: ☒

COURSE DESCRIPTION

In this course, students will discuss the theoretical and practical issues related to the application of enterprise systems within organizations. The main focus of this course is to demonstrate how enterprise systems integrate information and organizational processes across functional areas with a unified system comprised of a single database and shared reporting tools.

Date: June 19, 2020

Prepared By: Nathan Abourbih

Approved by: 

BRIAN LOBBAN
Dean, School Of Business And Information Technology,
Creative Arts, Design, Music And Hospitality

Effective: Fall 2020, Winter 2021, Spring 2021

RELATIONSHIP TO PROGRAM VOCATIONAL LEARNING OUTCOMES

PROGRAM LEVEL	
This course contributes to your program by allowing you to demonstrate the following vocational learning outcomes:	
Program(s)	Vocational Learning Outcomes
I.T. Business Analysis	College Standards 1. Analyze the information collected during the requirements elicitation to identify gaps and define the capabilities of an acceptable solution using appropriate methods, tools, techniques and documentation. 2. Apply negotiation and problem-solving skills to build effective stakeholder alignment.
I.T. Business Analysis - Hanson	Ministry Standards 1. Analyze the information collected during the requirements elicitation to identify gaps and define the capabilities of an acceptable solution using appropriate methods, tools, techniques and documentation.

COURSE CURRICULUM

Topics/Concepts Covered in This Course

- Introduction to Enterprise Systems for Management
- Systems Integration
- Enterprise Systems Architecture
- Development Life Cycle
- Implementation Strategies
- Software and Vendor Selection
- Operations and Post Implementation
- Organizational Change and Business Process Reengineering
- Global, Ethics, and Security
- Supply Chain Management
- Customer Relationship Management
- ERP Software

COURSE LEVEL: Learning Outcomes and Objectives	
To earn credit for this course, you must reliably demonstrate your ability to:	
Learning Outcome	Objectives
1. Discuss the evolution of enterprise information systems in organizations.	1.1 Describe the information systems evolution and its historical role in organizations leading to systems integration and eventually Enterprise Resource Planning (ERP). 1.2 Describe ERP system components and their architecture.

Learning Outcome	Objectives
	1.3 List the benefits and drawbacks of implementing ERP systems, as well as describe how they help organizations improve efficiencies and worker productivity. 1.4 Describe the implementation process of ERPs including: the life cycle, business process reengineering, project management, and change management. 1.4.1 Describe staff, vendor, consultant and organization roles in successful implementation processes. 1.5 Investigate the ethical, global and security challenges while implementing an ERP system. 1.6 Research ERP vendors and industry trends.
2. Discuss how system integration is heavily impacted by organizational structure and its information system.	2.1 Describe how organizational structures impact information systems. 2.2 Identify types of functional silos in organizations. 2.3 Describe how technology generations and architectures have influenced information systems silos in organizations. 2.4 Describe what systems integration is and why it's important for organizations. 2.5 Defend the role of enterprise resource planning (ERP) systems in achieving systems integration.
3. Describe in detail the enterprise systems modules and architectures.	3.1 Examine in detail the enterprise systems modules and architecture. 3.2 Model a well-designed architecture of an ERP implementation. 3.3 List various types of ERP architectures and the related benefits and drawbacks of each architecture. 3.4 Describe service-oriented architecture and its impact on ERP systems. 3.5 Describe cloud architecture and its impact on ERP systems.
4. Contrast various ERP implementation strategies and issues affected by the development life cycle that support businesses.	4.1 Describe the systems development life cycle (SDLC). 4.2 Identify the problems and alternatives with SDLC. 4.3 Describe the key issues in ERP implementation strategy. 4.4 Describe the ERP implementation life cycle. 4.5 Describe the rapid implementation methodologies. 4.6 Compare and contrast SDLC and ERP life cycles. 4.7 Describe the role of top management, consultants, and subject matter experts (SMEs) in the ERP life cycle. 4.8 Relate the importance of project management office (PMO) and the project organization to a successful ERP implementation.

Learning Outcome	Objectives
	4.9 List and describe the components of a project organization and the roles and responsibilities of each.
5. Explain various implementation platform components and describe how they work together to support business.	5.1 Describe the ERP components and how they work together to support business. 5.2 Explain why third-party products are needed to operationally round out ERP system functionality and the issues involved in using them. 5.3 Identify the impacts of an ERP implementation on platform components such as data security, system reliability, and sustainability. 5.4 Compare differences between implementations that require minimal or no system modifications, against those that require custom system modifications. 5.4.1 Identify short-term and long-term impacts on the system and company.
6. Research several software and vendor ERP systems and describe the steps for a successful purchase and implementation of an ERP system.	6.1 Identify initial steps in the process for the successful purchase and implementation of an ERP system. 6.2 Determine the total cost of ownership and what it is to partner with an ERP vendor. 6.3 Describe how the purchase phase of the ERP is critical to the adoption of change management process. 6.4 Identify the steps involved in negotiating a contract with a vendor.
7. Describe the process of transferring knowledge to operational staff and it's importance to long-term ERP system success.	7.1 Describe all the components to a successful "Go-live" and determine their readiness. 7.2 Describe what's involved in stabilizing the system after "Go-live" and how to track and address problems and issues on a daily basis. 7.3 Describe the transition from developing a system to supporting it in a production environment. 7.4 Describe the process of transferring knowledge to operational staff and its importance to long-term system success. 7.5 Define training practices before and after "Go-live".
8. Rationalize the purpose for organizational change and business process reengineering, to create a competitive advantage.	8.1 Explain the relationship between ERP system adoption and business process reengineering (BPR) best practices. 8.2 Discuss the commitment role that senior management take to endorse and assist in overcoming resistance to change in business processes to meet the company vision and goals. 8.3 Develop an awareness of Organizational Project Management Maturity Model (OPM3) and how it is used to assess an organization's ability to implement and ERP system successfully.

Learning Outcome	Objectives
	8.4 Introduce business process management (BPM) and discuss its relationship with BPR.
9. Explain the ethical, legal and security issues related to ERP systems and their components.	9.1 Describe the business and cultural implications of outsourcing and offshoring. 8.1.1 Describe the Software as a Service (SaaS) model. 9.2 Identify ethical and legal issues related to ERP systems, its implementation, and how to protect company assets. 9.3 List various components to system security. 8.1.3 Describe why security must be planned, tested, and ready by the time the ERP implementation is at Go-live. 9.4 Describe the ERP's role in green IT. 9.5 Examine the impact of compliance and of laws such as the Sarbanes-Oxley Act on ERP implementations.
10. Describe supply chain components and different levels of supply chain integration.	10.1 Describe the supply chain network and management drivers. 10.2 Investigate the complexity and importance of the integration of supply chain. 10.3 Identify supply chain components, processes, and flows. 10.4 List and describe the different levels of supply chain integration. 10.5 Track and report the impact of ERP on supply chain management.
11. Identify components Customer Relation Management (CRM) systems and its relationship with other enterprise software.	11.1 Describe the customer relationship process. 11.2 Define the evolution, current status, and categories of the customer relationship management (CRM) system. 11.3 List the components and architecture of CRM systems. 11.4 Follow the CRM life cycle and compare its relationship with other enterprise software. 11.5 Examine and measure the impact of CRM on an organization.
12. Use ERP software and features to simulate the development and implementation of a technical solutions to effectively support organizations.	12.1 Explore customer relationship management (CRM) features of ERP software. 12.2 Simulate supply chain management (SCM) elements within ERP software. 12.3 Investigate ERP tools and features that integrate reporting information across marketing, sales, accounting and HR business areas to obtain a unified competitive business.

Essential Employability Skills

Communication

- communicate clearly in written, spoken, and visual form that fulfills purpose/needs of audience.
- respond to written, spoken, or visual messages in a manner that ensures effective communication.

Numeracy

- not applicable

Critical Thinking and Problem Solving

- apply a systematic approach to solve problems.
- use a variety of thinking skills to anticipate and solve problems.

Information Management

- locate, select, organize, and document information using appropriate technology and info systems.
- analyze, evaluate, and apply relevant information from a variety of sources.

Interpersonal

- show respect for the diverse opinions, values, belief systems, and contributions of others.
- interact with others in groups in ways that contribute to effective working relationships.

Personal

- manage the use of time and other resources to complete projects.
- take responsibility for one's own actions, decisions, and consequences.

Delivery Method

- Classroom: Course is delivered through scheduled synchronous teaching that may be face-to-face and/or virtual.
- Online: Course is fully delivered through asynchronous teaching.
- Hybrid: Course combines scheduled synchronous and unscheduled asynchronous teaching.
- HyFlex: Course includes both synchronous and asynchronous learning and the student can move between both components seamlessly.

Learning Activities

- Lectures
- Class Discussions
- Labs
- Group Work
- Research
- Guest Speaker(s)
- In-Class Exercises
- Case Studies

Resources Required

Books

Motiwalla, Luvai; Thompson, Jeffrey, *Enterprise Systems for Management*, 2/e, 2nd, Prentice Hall
ISBN: 978-0-13-214576-3

Evaluation Plan

Grading Scheme

A	80% - 100%
B	70% - 79%
C	60% - 69%
D	50% - 59%
F	0% - 49%

Evaluation Method	Value (%)
Assignments	40%
Applied Activities(G)	40%
Quizzes/Test/Exam	20%

ADDITIONAL INFORMATION

College

Academic Policies

Students must adhere to all academic policies. These are available on myCambrian as well as the College website at <https://cambriancollege.ca/about/official-documents-and-policies/academic-policies/>.

Prior Learning Assessment and Recognition (PLAR)

Students may contact the Pathways office at pathways@cambriancollege.ca

Transfer Credit

Students wishing to have courses from other programs or institutions assessed for equivalency and/or transfer credit should contact transfercredits@cambriancollege.ca.

Accessibility Services

Cambrian is committed to creating an inclusive learning environment where we support our students' success. Students who are registered with the Glen Crombie Centre can access accessibility and/or counselling services by:

1. Registering and scheduling an appointment through Clockwork in the Student Portal in myCambrian.
2. Contacting 705-566-8101 ext. 7420 or 7311, or
3. E-mailing accessibilityservices@cambriancollege.ca

Testing Practices at Cambrian

Many courses include major tests and/or final exams. The practice at Cambrian requires that these types of test situations involve proctoring to ensure academic integrity. Online tests/exams may employ a proctoring service to enable you to take your exam from a location of your choosing within a period specified by your instructor. When you are taking an online test/exam, the proctoring service may capture your video, screen, audio and web surfing particulars to protect academic integrity. Cambrian College collects, uses, discloses and retains personal information in compliance with the Freedom of Information and Protection of Privacy Act (FIPPA). Your personal information is being collected under the authority of the Ontario Colleges of Applied Arts and Technology Act S.O. 2002, c.8, Sched. F. This information will be used for the purpose of administering a test/exam through an online proctoring service acting as an authorized agent of the College. Please refer to Cambrian's Confidentiality of Student Records Policy for more details. If you have any questions regarding the collection of your personal information, please contact Vice President Academic, Cambrian College, 1400 Barrydowne Rd., Sudbury ON P3A 3V8, 1-705-566-8101.

Wabnode Centre for Indigenous Services

Students are encouraged to come to Wabnode to meet our Elders and to access personal counselling or academic tutoring. Our team will also be able to provide students with information related to funding and scholarship and bursary opportunities for Indigenous students. We encourage students to come to Wabnode for cultural, sports and community activities. Students can also contact Wabnode at wabnode@cambriancollege.ca

Copyright

Students may not submit work or partake in activities that infringe on the Canadian Copyright Act, or the related Cambrian College Fair Dealing Guidelines and Copying Guidelines. The guidelines may be viewed at: <http://cambriancollege.libguides.com/copyrightinfo> or by contacting the Library for questions related to copyright: library@cambriancollege.ca

Out-of-Class Assistance

The faculty will inform students of their Out-of-Class availability through the Learning Management System site (Moodle).

Audio/Visual Capture

Sounds and images from this class, and contributions made by a participant, virtually or in-person, are recorded under the authority of the Ontario Colleges of Applied Arts and Technology Act, 2002. The main purpose for these recordings is to allow students enrolled in this course, whether they attend the particular class in person, virtually, or at all, to review class content and activity.

These recordings may also be reviewed by faculty members to prepare future classes, to evaluate students, to collaborate in program/course development/review, or to provide feedback to students or other faculty. Any questions about this collection may be addressed to your respective Dean.

Equity, Diversity and Inclusivity

Cambrian is committed to building and preserving an equitable, diverse, and inclusive learning community where students, faculty, and staff may achieve their full potential in an environment characterized by equality of respect and opportunity. All students and employees have the right to live and work in an environment that is free from discrimination and harassment. Therefore, Cambrian College will not tolerate any form of discrimination or harassment in its employment, education, accommodation, or business dealings. For more information, please visit: <https://cambriancollege.ca/about/official-documents-and-policies/equity-human-rights-and-accessibility/>

Tests and Evaluations

Students are required to write tests and/or complete evaluations as scheduled. Exceptions may be made in the event of an emergency or a sanctioned event (e.g., varsity sports, field trip, religious observances, etc.). In non-emergency situations, students are expected to contact the faculty member in the event that they can not be present for a test/evaluation to explain the reason for their absence. At the discretion of the faculty member, an alternate time for the test/evaluation may be allowed.

Attendance

Regular attendance is strongly encouraged as it contributes to student success. Some courses have specific lab/participation requirements. Students are expected to be aware of, and adhere, to these requirements.

Modifications to the Course Evaluation Plan

Under exceptional circumstances, Cambrian reserves the right to alter the Course Evaluation Plan. Upon approval from the Dean, faculty members will notify students of any modification to the Course Evaluation Plan and post to the course Learning Management System site (Moodle).

School of Business and Information Technology, Creative Arts, Design, Music and Hospitality Regulations**Lab Attendance**

Lab attendance is mandatory since lab grades include an evaluation of lab reports and lab performance. Lab performance can only be judged in person, requiring that students attend all scheduled labs.